

Step4d_CFC and CFC_plot Usage Guide

Step4d_CFC

Overview

Step4d_CFC calculates the mean vector length (MVL) between regions pairs for a participant. The function utilizes the beamformed data from Step3. The calculation is performed across all trials and specified region pairs. If no regions are specified for high or low frequency, the function defaults to using all regions in the data. A .mat file is saved containing MVL values by low frequency ROIs x high frequency ROIs. See Appendix A for a flowchart describing the function, and Appendix B for more information on CFC and MVL.

Usage

The function requires participant ID, visit number and a config structure. The config structure must also include “.meta” path names (or the directory in which the data files are stored in, specifically cfg.meta.project_path and cfg.meta.analysis_name). A sample config structure and function call are shown below:

```
config.step4d.lowFreqband = [8 12];
config.step4d.highFreqband = [30 70];
config.step4d.chanhigh = {'L_V8', 'L_VVC', 'L_PIT'};
config.step4d.chanlow = {'L_VMV1', 'L_VMV2'};
Step4d_CFC(config, "WMP_19", 1);
```

Limitations/Things to Note

1. The value is an average across trials for each region pair.
2. Regions can be specified for low and/or high frequency or neither. If it is not specified, all labels are used.
3. The function checks if the given ROI names exist in the data that is being used. If it does not, the function will stop running and display an error indicating which names it could not find.
4. Fieldtrip has a CFC function that may be worth digging more into: ft_crossfrequencyanalysis. The MVL function used in Step4d_CFC is an adaptation of a sub-function contained in ft_crossfrequencyanalysis. The main difference is Step4d_CFC uses the analytic signal, instead of the fourier spectrum.

CFC_plot

Overview

CFC_plot utilizes the output of Step4d_CFC to plot average low frequency ROI x high frequency ROI heatmaps of the MVL values for a given set of participants and visits. The function may be used with just one participant ID. The function saves a .mat file containing the averaged MVL values, and the participant IDs and visit numbers used to calculate that average. The heatmaps and .mat files are stored in a group folder if more than one participant ID is given. If only one participant ID is given, it will save in the corresponding session number if only one is given, or in the participant folder if several visit numbers are given.

Usage

The function requires a config structure as an input. A sample config structure and function call is shown below. The config structure must also include “.meta” path names, as described above. In this config structure, the average is taken for session 1 for participant 1, and sessions 1 and 2 for participant 4. It will take the average of the MVL values between 8-12 Hz and 30-70 Hz.

```
config.CFCplot.pids = ["WMP_01" "WMP_04"];  
config.CFCplot.visitnum = {[1] [1 2]};  
config.CFCplot.lowFreqband = [8 12];  
config.CFCplot.highFreqband = [30 70];  
CFC_plot(config);
```

Sample Output:

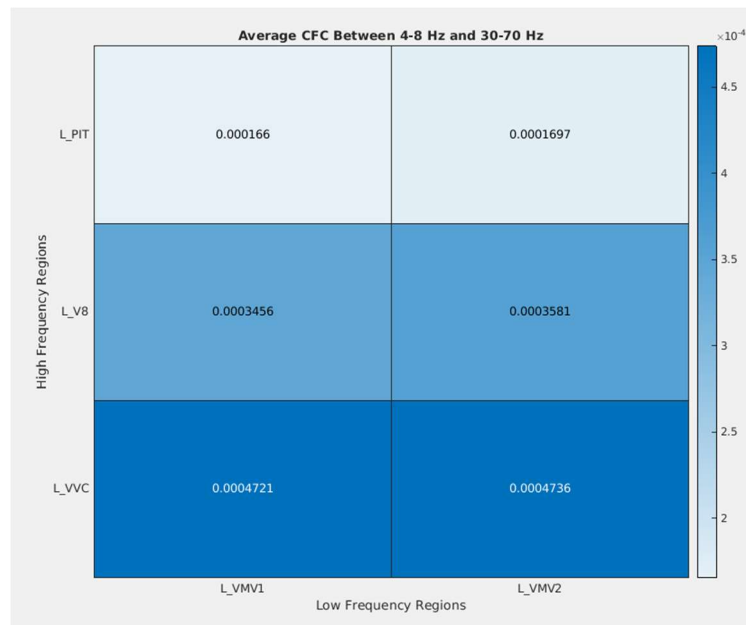


Figure 1. Sample CFC heatmap where lowchan and highchan were specified for data

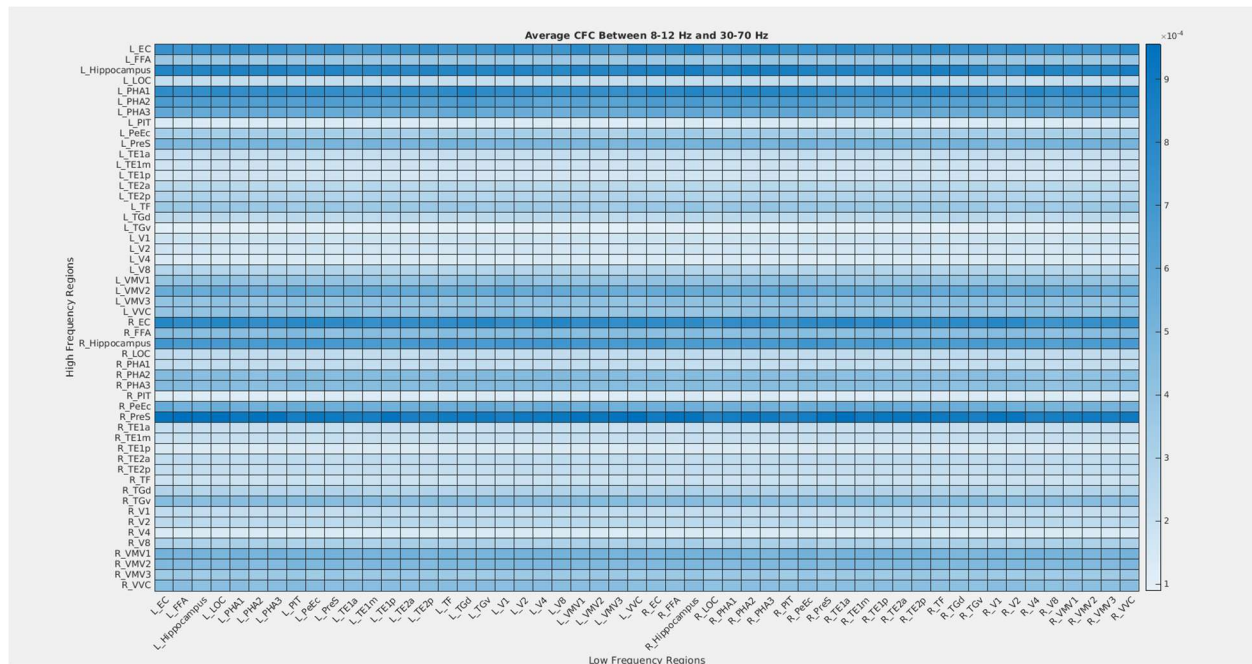
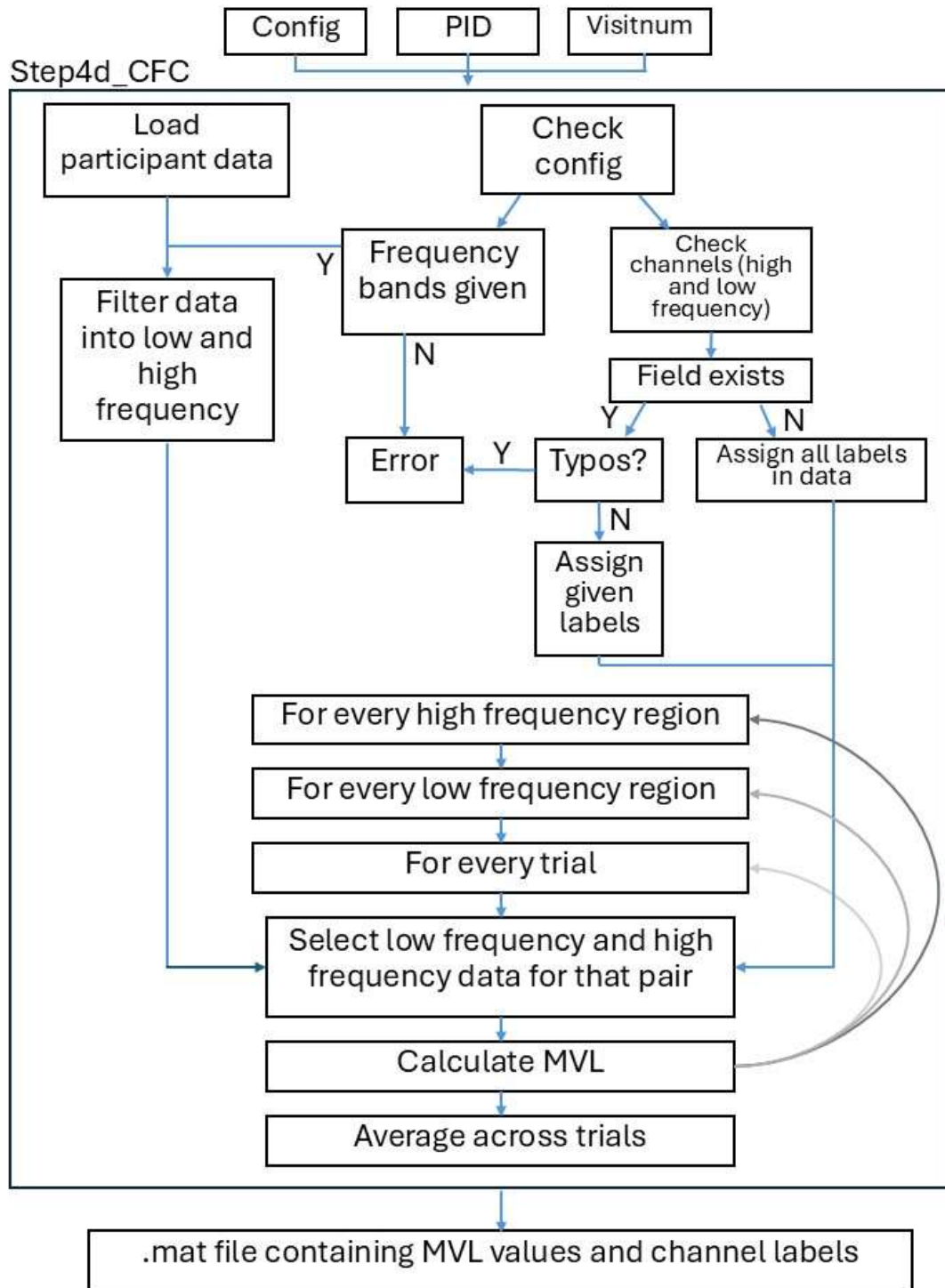


Figure 2. Sample CFC heatmap where lowchan and highchan were not specified, and function used all ROIs by default

Limitations/Things to Note

1. The function assumes that the data for all participants and visits use the same ROIs in low and high frequency.
2. This matrix is not symmetric.

Appendix A: Step4d_CFC Flowchart



Appendix B: Useful Papers

Hülsemann, M. J., Naumann, E., & Rasch, B. (2019). Quantification of Phase-Amplitude Coupling in Neuronal Oscillations: Comparison of Phase-Locking Value, Mean Vector Length, Modulation Index, and Generalized-Linear-Modeling-Cross-Frequency-Coupling. *Frontiers in Neuroscience*, 13. <https://doi.org/10.3389/fnins.2019.00573>

Siebenhühner, F., Wang, S. H., Arnulfo, G., Lampinen, A., Nobili, L., Palva, J. M., & Palva, S. (2020). Genuine cross-frequency coupling networks in human resting-state electrophysiological recordings. *PLOS Biology*, 18(5), e3000685. <https://doi.org/10.1371/journal.pbio.3000685>